

Paper Decoder

September 2026 compilation

3 days · 9 stories

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1. India launches commercial production of bio-based biodegradable Polylactic Acid plastic

The initiative under the Biotechnology for Economy, Environment and Employment (BioE3) framework will establish a pilot facility in Uttar Pradesh to develop advanced bioplastics.

What happened	The government launched the commercial use of bio-based biodegradable Polylactic Acid (PLA) plastic under the Biotechnology for Economy, Environment and Employment (BioE3) framework.
Why UPSC cares	This launch under the BioE3 (Biotechnology for Economy, Environment and Employment) framework is crucial for studying sustainable manufacturing and India's efforts to reduce plastic pollution through green alternatives.
Syllabus	GS3 Environment: Conservation, environmental pollution and degradation
Chapter	Environment: Pollution: Types, Sources & Control

Read it for

1. Understand the science of Polylactic Acid (PLA) as a biodegradable polymer.
2. Examine the goals of the Biotechnology for Economy, Environment and Employment (BioE3) framework.
3. Analyze the growth and potential of India's bioeconomy sector.

Key terms

Polylactic Acid (PLA)

A biodegradable and bioactive thermoplastic aliphatic polyester derived from renewable resources like corn starch or sugarcane.

BioE3 Framework

A government initiative standing for Biotechnology for Economy, Environment and Employment, aimed at fostering high-performance biomanufacturing.

Bioeconomy

An economic model based on the utilization of biological resources, biotechnology, and bioprocesses to produce goods, services, and energy.

Biodegradable Plastic

Plastics that can be decomposed by the action of living organisms, usually microbes, into water, carbon dioxide, and biomass under specific conditions.

Prelims: how it could come

Consider the following statements regarding Polylactic Acid (PLA):

1. It is a bio-based polymer typically synthesized from renewable agricultural resources like sugarcane or corn starch.
2. All PLA-based plastics degrade rapidly in standard backyard home composting bins without requiring industrial composting facilities.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Answer: (a) Statement 1 is correct: Polylactic Acid (PLA) is a biodegradable polyester derived from renewable biomass sources, such as fermented plant starch (usually corn, cassava, sugarcane or sugar beet pulp). Statement 2 is incorrect: Although PLA is biodegradable, it requires specific high-temperature industrial composting conditions (around 58-60 degrees Celsius with controlled humidity and microbes) to degrade efficiently. It does not decompose rapidly in standard backyard home composting environments.

Mains: how it could come

Explain the significance of the Biotechnology for Economy, Environment and Employment (BioE3) framework in driving India's bioeconomy. Discuss the challenges associated with the commercial scaling of biodegradable plastics like Polylactic Acid (PLA). (150 words, 10 marks)

The BioE3 framework aims to transition India towards high-performance biomanufacturing, promoting sustainable alternatives like Polylactic Acid (PLA) to mitigate environmental degradation.

- Significance of BioE3: Fosters high-value biomanufacturing, reduces carbon footprint, integrates industry-academia collaboration, and generates employment.
- Economic Growth: Supports India's rapidly expanding bioeconomy, which grew from 10 billion dollars in 2014 to nearly 195 billion dollars.
- Challenges of PLA Scaling: High cost of production compared to conventional petroleum-based plastics.
- Infrastructure Gaps: Requires specialized industrial composting infrastructure, as PLA does not degrade quickly in normal landfills or oceans.

To successfully scale bioplastics, India must couple production incentives with waste management policies that mandate industrial composting facilities.

Keywords to use: BioE3 framework, Polylactic Acid (PLA), Bioeconomy, Industrial composting, Biomanufacturing

GS2 · International Relations

2. G4 nations push for urgent United Nations Security Council reform

Foreign Ministers of India, Brazil, Germany and Japan met in New York to press for structural expansion of the United Nations Security Council.

What happened	Foreign Ministers of G4 countries met on the sidelines of the United Nations General Assembly in New York to demand urgent reform and expansion of the United Nations Security Council.
Why UPSC cares	The joint call by Group of Four (G4) nations (India, Brazil, Germany, Japan) highlights the urgent need for United Nations Security Council (UNSC) reforms, a key topic for Mains on global governance.
Syllabus	GS2 International Relations: Bilateral, regional and global groupings and agreements involving India and affecting India's interests.
Chapter	Polity: India's External Relations

Read it for

1. Understand the composition and core objectives of the G4 grouping.
2. Examine the structural flaws of the United Nations Security Council.
3. Analyze the role of the Intergovernmental Negotiations framework.

Key terms

G4 Nations

A coalition comprising India, Brazil, Germany and Japan that support each other's bids for permanent seats on the United Nations Security Council.

United Nations Security Council

The principal crisis management body of the United Nations tasked with maintaining international peace and security.

Inter-governmental Negotiations

A framework established under the United Nations General Assembly to discuss reforms of the United Nations Security Council.

Veto Power

The special voting power held by the five permanent members of the United Nations Security Council to block any substantive resolution.

Mains: how it could come

Analyze the hurdles in reforming the United Nations Security Council and discuss India's strategic rationale for seeking a permanent seat in the Council. (150 words, 10 marks)

The United Nations Security Council reflects the geopolitical realities of 1945, making its structure outdated and unrepresentative of contemporary global power dynamics.

- Geopolitical resistance from existing permanent members who are reluctant to dilute their veto powers or share global authority.
- Regional rivalries where neighboring countries oppose each other's inclusion, such as the Coffee Club opposing the G4 bids.
- India's credentials as the world's largest democracy, a major contributor to United Nations peacekeeping missions, and an emerging economic powerhouse.
- The lack of representation from Africa and Latin America, weakening the moral legitimacy and efficacy of multilateral decisions.

India must build broader coalitions with developing nations and leverage multilateral platforms to push for text-based negotiations in the United Nations General Assembly.

Keywords to use: Multilateralism, Veto Power, Inter-governmental Negotiations, Coffee Club, Equitable Representation

GS3 · Economy

3. Government cuts basic customs duty on crude edible oils to cool prices

The reduction in import duties on crude sunflower, soybean and palm oils aims to provide relief to consumers facing high inflation.

What happened	The Government reduced the Basic Customs Duty on crude sunflower, soybean and palm oils to moderate rising domestic food prices.
Why UPSC cares	The reduction in Basic Customs Duty (BCD) on crude edible oils is a key fiscal measure to control domestic inflation, illustrating the trade-offs between consumer relief and domestic oilseed production incentives.
Syllabus	GS3 Indian Economy: Inclusive growth and issues arising from it
Chapter	Economy: External Sector of India

Read it for

1. Understand how import duty adjustments act as a fiscal tool to manage domestic inflation.
2. Examine the rationale behind maintaining a duty differential between crude and refined oils.
3. Analyze the impact of global price trends on domestic food security and consumer welfare.

Key terms

Basic Customs Duty

It is a tax levied on goods imported into India, calculated as a percentage of the value of the goods. It serves as a tool for revenue generation and domestic industry protection.

Import Duty Differential

The gap between the duty rates applied on crude raw materials and finished refined products. Maintaining this gap protects domestic refining industries and encourages value addition.

Headline Inflation

The total inflation within an economy, including commodities such as food and energy prices, which tend to be much more volatile and prone to sudden spikes.

Landed Cost

The total cost of a product once it has arrived at the buyer's door, including the purchase price, freight, insurance, and all applicable customs duties and taxes.

Prelims: how it could come

Consider the following statements regarding import duties and domestic price management in India:

- 1. Reducing the Basic Customs Duty on raw material imports lowers the landed cost for domestic processing industries.**
- 2. Maintaining an import duty differential between crude and refined oils is primarily aimed at discouraging domestic value addition.**
- 3. International commodity price surges can directly transmit into domestic food inflation if trade policy buffers are not utilized.**

Which of the statements given above are correct?

- a. 1 and 2 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Answer: (c) Reducing import duties lowers the landed cost of raw materials, helping processors pass benefits to consumers. Duty differentials are maintained to protect domestic refiners and encourage domestic processing rather than discouraging it. Unchecked international spikes easily transmit to local markets, making duty cuts a necessary buffer.

Mains: how it could come

Analyze the structural challenges facing India's edible oil sector and discuss the role of trade policy measures, such as customs duty adjustments, in managing domestic food inflation. (150 words, 10 marks)

India imports a significant portion of its edible oil requirements, making domestic prices vulnerable to international market shocks and necessitating timely fiscal interventions.

- High import dependency exposes the domestic market to global price volatility, supply chain disruptions, and currency fluctuations.
- Tariff adjustments like reducing Basic Customs Duty act as immediate shock absorbers to lower landed costs and contain food inflation.
- Maintaining a duty differential between crude and refined oils protects domestic refining capacity and prevents dumping of finished goods.
- Long-term solutions require boosting domestic oilseed productivity through missions like the National Mission on Edible Oils.

A balanced approach combining short-term tariff rationalization with long-term domestic production enhancement is essential for sustainable edible oil security.

Keywords to use: Import dependency, Basic Customs Duty, Food inflation, Duty differential, Oilseed productivity

1. Centre extends AFSPA in parts of Manipur, Nagaland and Arunachal

The Ministry of Home Affairs has extended the Armed Forces (Special Powers) Act, 1958, across designated disturbed areas in three north-eastern states for six months.

What happened	The Ministry of Home Affairs extended AFSPA for six months from 1 October in parts of Manipur, Nagaland, and Arunachal Pradesh.
Why UPSC cares	This extension of the Armed Forces (Special Powers) Act (AFSPA) highlights ongoing security challenges in the Northeast. It is crucial for understanding internal security and federal relations under General Studies Paper 3.
Syllabus	GS3 Internal Security: Linkages between development and spread of extremism, role of external state and non-state actors in creating challenges to internal security.
Chapter	Polity: National Integration: Key Facts & Functions

Read it for

1. Statutory procedure for declaring an area as disturbed under Section 3 of AFSPA.
2. Special operational powers and legal protections conferred on armed forces personnel.
3. Security and human rights debates surrounding the phased withdrawal of AFSPA in the Northeast.

Key terms

Armed Forces (Special Powers) Act, 1958 (AFSPA)

A Parliamentary enactment that grants special powers to the armed forces to maintain public order in designated disturbed areas.

Disturbed Area

An area notified under Section 3 of AFSPA where the use of armed forces in aid of civil power is deemed necessary due to civil unrest or insurgency.

Prior Sanction

A statutory requirement under Section 6 of AFSPA where legal proceedings against military personnel acting under the law require prior central approval.

Aid to Civil Power

The deployment of military or central paramilitary forces to support civilian administration and state police in quelling unrest and restoring public order.

Prelims: how it could come

With reference to the Armed Forces (Special Powers) Act, 1958 (AFSPA), consider the following statements:

1. An area can be declared as a 'disturbed area' either by the Central Government or by the Governor of the concerned State.
2. Armed forces personnel operating in a disturbed area have the legal authority to set up summary courts to prosecute arrested persons.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Answer: (a) Statement 1 is correct: Under Section 3 of AFSPA, the Central Government, the Governor of the State, or the Administrator of the Union Territory can declare the whole or any part of a State or Union Territory as a disturbed area. Statement 2 is incorrect: AFSPA gives armed forces personnel powers of search, seizure, and arrest without warrant, but Section 5 explicitly mandates that any person arrested must be handed over to the nearest police station with the least possible delay. The armed forces have no power to set up courts or try civilians.

Mains: how it could come

Examine the operational necessity and human rights concerns associated with the enforcement of the Armed Forces (Special Powers) Act, 1958 in the North-Eastern region. (150 words, 10 marks)

The Armed Forces (Special Powers) Act, 1958 grants sweeping powers to military personnel deployed in designated disturbed areas to assist civil administration in counter-insurgency operations.

- Operational necessity: Porous international borders, complex ethnic fault lines, and heavily armed insurgent groups require robust deterrence and specialized military capabilities beyond the capacity of local police.
- Human rights and legal concerns: Wide powers under Section 4, including use of lethal force and warrantless arrests, coupled with Section 6 immunity requiring Central sanction for prosecution, create accountability gaps.
- Democratic deficit: Prolonged reliance on extraordinary military laws normalizes an emergency regime and risks alienating local populations, undermining civic integration.
- Institutional reviews: The Justice Jeevan Reddy Committee recommended repealing AFSPA and incorporating necessary security provisions into ordinary criminal law to avoid stigmatization.

A progressive phased withdrawal, continuous capacity enhancement of state police forces, and adherence to the Supreme Court guidelines in the Naga People's Movement for Human Rights case are vital.

Keywords to use: disturbed areas, Section 6 immunity, Justice Jeevan Reddy Committee, police modernization, counter-insurgency

GS3 · Environment

2. Nepal PM proposes Himalayan climate resilience mechanism with India and China

Nepal Prime Minister Balendra Shah proposed a joint mechanism to share satellite data and early-warning information following devastating flash floods in the region.

What happened	Nepal proposed a trilateral Himalayan Climate Resilience Mechanism with India and China at the 81st United Nations General Assembly (UNGA) session following devastating flash floods that killed 1,452 people.
Why UPSC cares	The proposed Himalayan Climate Resilience Mechanism is vital for regional ecological security. It is highly relevant for transboundary climate cooperation and disaster management under General Studies Paper 3.
Syllabus	GS3 Environment: Environmental pollution and degradation, disaster and disaster management.
Chapter	Environment: Consequences of a Warming World

Read it for

1. Understand the impact of climate change on the fragile Himalayan ecosystem, specifically focusing on permafrost melting and ice-rock avalanches.
2. Examine the role of transboundary cooperation in disaster management and early warning systems among India, Nepal and China.
3. Analyze the significance of multilateral platforms like the United Nations General Assembly (UNGA) in addressing regional climate vulnerabilities.

Key terms

Permafrost

Ground that remains completely frozen at or below zero degrees Celsius for at least two consecutive years, commonly found in high-latitude or high-altitude regions.

Glacial Lake Outburst Flood (GLOF)

A sudden release of water from a lake fed by glacier melt, often triggered by avalanches, landslides or the collapse of the natural dam holding the water.

Cascading Disasters

An extreme event where one primary hazard triggers a sequence of secondary hazards, leading to compounded impacts that are harder to manage.

Climate Resilience

The capacity of social, economic and environmental systems to cope with a hazardous event, trend or disturbance by responding or reorganising in ways that maintain their essential function.

Prelims: how it could come

With reference to permafrost, consider the following statements:

1. It is defined as ground that remains completely frozen at or below zero degrees Celsius for at least two consecutive years.
2. The thawing of permafrost can release greenhouse gases like methane and carbon dioxide, creating a positive feedback loop in global warming.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Answer: (c) Statement 1 is correct. Permafrost is defined as ground (soil, rock, and organic material) that remains at or below zero degrees Celsius for at least two consecutive years. Statement 2 is correct. As permafrost thaws, organic matter trapped in the ice begins to decay, releasing large amounts of methane and carbon dioxide, which are potent greenhouse gases. This amplifies global warming, creating a positive feedback loop.

Mains: how it could come

The fragile ecosystem of the Hindu Kush Himalayan region is increasingly threatened by cascading disasters driven by climate change. In this context, evaluate the ecological and geopolitical challenges in establishing a transboundary disaster resilience mechanism among India, Nepal, and China. (150 words, 10 marks)

The Hindu Kush Himalayan region is highly vulnerable to warming, leading to accelerated glacial melt, permafrost degradation, and frequent transboundary disasters like flash floods.

- **Ecological Vulnerability:** High altitude warming causes ice-rock avalanches, landslides, and Glacial Lake Outburst Floods (GLOFs) that cascade downstream across national borders.
- **Geopolitical Friction:** Strategic mistrust and border disputes between India and China hamper seamless satellite data sharing, joint technical research, and real-time early warning coordination.
- **Institutional Gaps:** Existing bilateral frameworks are fragmented and lack a unified, legally binding trilateral mechanism to manage shared river basins and mountain ecosystems.
- **Socio-Economic Impact:** Downstream populations in Nepal and India suffer immense loss of lives, livelihoods, and critical infrastructure due to uncoordinated disaster responses.

Establishing a science-first, non-political platform under regional bodies or United Nations guidance can foster trust, enabling collaborative data sharing and climate-resilient infrastructure development.

Keywords to use: Hindu Kush Himalaya, Glacial Lake Outburst Flood (GLOF), transboundary cooperation, climate resilience, geopolitical mistrust

GS3 · Science and Technology

3. Artificial intelligence could boost long term aggregate labor productivity by 3.8 percent

A new research paper analyzing patent and employment data shows that artificial intelligence technologies can significantly enhance output per worker when complementing skilled labor.

What happened	A study analyzing patent data from 2000 to 2017 in developed economies indicates that Artificial Intelligence (AI) can increase aggregate labor productivity by up to 3.8 percent over the long term.
Why UPSC cares	This International Monetary Fund (IMF) study on Artificial Intelligence (AI) and productivity is key for understanding the economic impact of technology and future of work under General Studies Paper 3.
Syllabus	GS3 Science and Technology: developments and their applications and effects in everyday life.
Chapter	Science & Tech: AI & Emerging Digital Tech

Read it for

1. Impact of Artificial Intelligence (AI) on aggregate labor productivity and economic growth.
2. The complementary relationship between advanced technology and skilled labor force.
3. Role of patent activity as a metric for measuring technological innovation and economic returns.

Key terms

Labor Productivity

The measure of economic output produced per unit of labor input, typically calculated as real gross domestic product per hour worked or per worker.

Capital-Complementing Technology

Technologies that increase the productivity of workers, allowing them to produce more output, rather than replacing them entirely.

Aggregate Output

The total value of all final goods and services produced in an economy over a specific period of time.

Patent Activity

The filing and granting of patents, often used as an indicator of technological innovation and research and development intensity in an economy.

Prelims: how it could come

With reference to economic growth and labor productivity, consider the following statements:

1. Labor productivity in an economy is solely determined by the amount of physical capital per worker, and is independent of technological advancements.
2. When new technology complements skilled labor, it typically leads to higher aggregate economic returns compared to when it merely replaces existing tasks.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Answer: (b) Statement 1 is incorrect because labor productivity is heavily influenced by technological progress, human capital, and organizational efficiency, not just physical capital. Statement 2 is correct because technological innovations that complement human skills enhance worker efficiency and generate stronger, sustainable economic growth compared to simple automation that only substitutes labor.

Mains: how it could come

Discuss how Artificial Intelligence (AI) can act as a double-edged sword for developing economies like India. Suggest measures to ensure that AI adoption enhances aggregate labor productivity without exacerbating unemployment and economic inequality. (150 words, 10 marks)

Artificial Intelligence (AI) is a transformative technology capable of boosting aggregate labor productivity, but it also poses significant risks of labor displacement.

- **Productivity Boost:** AI can automate routine tasks, enabling workers to focus on high-value cognitive activities, thereby increasing output per worker.
- **Labor Market Challenges:** Developing nations face the risk of premature de-industrialization and job losses in service sectors like Information Technology (IT) and business process outsourcing.
- **Skill Mismatch:** The benefits of AI may disproportionately accrue to skilled workers, widening the wage gap and increasing economic inequality.
- **Complementarity Approach:** Focusing on AI as a tool to assist and enhance human capabilities rather than completely replacing workers.

India must invest in comprehensive upskilling programs, establish robust regulatory frameworks for ethical AI, and promote labor-complementing AI applications in agriculture and healthcare.

Keywords to use: labor productivity, labor displacement, upskilling, inclusive growth, complementarity

1. Russia backs India for a permanent seat in the UN Security Council

The Russian Foreign Minister has strongly advocated for India and Brazil to get permanent seats in an expanded United Nations Security Council.

What happened	Russia has expressed strong official support for India and Brazil to secure permanent seats in an expanded United Nations Security Council.
Why UPSC cares	Essential for GS2 International Relations. Covers major power alignments on United Nations Security Council reform and India's multilateral diplomacy.
Syllabus	GS2 International Relations: Bilateral, regional and global groupings and agreements involving India and/or affecting India's interests.
Chapter	Polity: India's External Relations

Read it for

1. Understand the composition and current structure of the United Nations Security Council.
2. Examine the arguments for and against reforming the global multilateral governance architecture.
3. Analyze the strategic geopolitical alignments regarding the Global South.

Key terms

United Nations Security Council

One of the principal organs of the United Nations tasked with maintaining international peace and security. It has fifteen members, including five permanent members with veto power.

Veto Power

The special voting power held by the five permanent members of the United Nations Security Council, allowing any one of them to block the adoption of a substantive resolution.

Global South

A term often used to refer to developing and lower-income countries located primarily in Asia, Africa, and Latin America, advocating for greater equity in global affairs.

BRICS

An intergovernmental organization comprising Brazil, Russia, India, China, South Africa, and newly added members, aimed at fostering economic and geopolitical cooperation.

Prelims: how it could come

Consider the following statements regarding the United Nations Security Council:

1. It consists of fifteen members in total, out of which ten are non-permanent members elected for two-year terms.
2. All permanent and non-permanent members possess veto power over substantive resolutions.
3. Regional representation in the Security Council is heavily criticized for underrepresenting the Global South.

Which of the statements given above is/are correct?

- a. 1 and 3 only
- b. 2 and 3 only
- c. 1 only
- d. 1, 2 and 3

Answer: (a) The United Nations Security Council has fifteen members: five permanent and ten non-permanent members elected for two-year terms, making statement 1 correct. Veto power is exclusively held by the five permanent members, making statement 2 incorrect. The current structure does not adequately represent Asia, Africa, and Latin America, leading to calls for reform from the Global South, making statement 3 correct.

Mains: how it could come

Analyze the structural anomalies of the United Nations Security Council and explain why India's inclusion as a permanent member is essential for a credible global governance architecture. (150 words, 10 marks)

The United Nations Security Council, created in nineteen forty-five, reflects the geopolitical realities of the post-World War II era rather than contemporary global demographics.

- Representation deficit: The council excludes entire regions like Africa and Latin America from permanent representation.
- India's credentials: India is the most populous nation, a major global economy, a vibrant democracy, and a leading contributor to United Nations peacekeeping missions.
- Multilateral legitimacy: Without restructuring, the council risks losing its moral authority and effectiveness in addressing twenty-first-century security threats.

India must continue forging strategic coalitions like the G4 and BRICS while building broader consensus among developing nations to push for comprehensive multilateral reforms.

Keywords to use: Multilateralism, Veto Power, Global South, Geopolitical Realities, Institutional Reform

GS3 · Science and Technology

2. India pitches for secure and democratic artificial intelligence at United Nations

External Affairs Minister S Jaishankar co-chaired an event on artificial intelligence and maritime security at the United Nations in New York.

What happened

External Affairs Minister Dr S Jaishankar co-chaired an event on artificial intelligence in international peace and security with an emphasis on maritime security at the

	United Nations.
Why UPSC cares	Crucial for GS3 Science and Technology. Connects artificial intelligence governance with international security and maritime domain awareness.
Syllabus	GS3 Science and Technology: Awareness in the fields of IT and Computers
Chapter	Science & Tech: AI & Emerging Digital Tech

Read it for

1. Note the dual nature of artificial intelligence in maritime security, spanning both domain awareness and cyber vulnerabilities.
2. Understand the governance challenges of autonomous maritime systems regarding attribution and accountability.
3. Examine the core principles of the New Delhi Declaration concerning democratic access to artificial intelligence.

Key terms

Maritime Domain Awareness

The effective understanding of anything associated with the maritime domain that could impact security, safety, economy, or environment.

Autonomous Maritime Systems

Vessels or systems operating in water that can make decisions and execute actions without direct human intervention using sensors and algorithms.

United Nations Convention on the Law of the Sea

The international agreement that establishes a legal framework for all marine and maritime activities across the globe.

Reconnaissance

Military observation of a region to locate an enemy or ascertain strategic features using advanced technological surveillance tools.

Mains: how it could come

Evaluate the strategic opportunities and security challenges posed by the integration of artificial intelligence into maritime security architectures. (150 words, 10 marks)

The convergence of artificial intelligence with maritime security presents a dual paradigm of enhanced situational awareness alongside complex geopolitical and cybersecurity risks.

- Enhances maritime domain awareness through predictive analytics, advanced reconnaissance, and efficient search and rescue operations.
- Secures critical infrastructure like ports and undersea communication cables against physical and digital disruptions.
- Introduces severe security risks, including autonomous weapon systems, compressed decision-making times, and escalation spirals.
- Complicates legal accountability and attribution, challenging existing frameworks of international law and state responsibility.

Global cooperation based on democratic principles, transparency, and adherence to international maritime law is essential to harness artificial intelligence safely.

Keywords to use: Maritime domain awareness, Autonomous systems, Attribution, Cyber operations, International law

GS3 · Environment & Ecology

3. Bandhavgarh Tiger Reserve hosts five day elephant rejuvenation camp in buffer zone

A five day elephant festival and health rejuvenation camp is underway near Parasi Gate in the Dhamokhar buffer zone of Bandhavgarh Tiger Reserve.

What happened	Bandhavgarh Tiger Reserve organised a five day rejuvenation camp for working elephants near Parasi Gate in its Dhamokhar buffer zone.
Why UPSC cares	Vital for GS3 Environment. Focuses on captive elephant welfare and management practices within protected tiger reserves in India.
Syllabus	GS3 Environment: Conservation, wildlife protection, and biodiversity management.
Chapter	Environment: Saving Species: Conservation in Action

Read it for

1. Role of working elephants in forest management and wildlife protection.
2. Buffer zone management and its importance in reducing human wildlife conflict.
3. Conservation initiatives for Asian elephants and community participation.

Key terms

Asian Elephant

The Asian Elephant (**Elephas maximus**) is listed as Endangered on the International Union for Conservation of Nature (IUCN) Red List and included in Schedule I of the Wildlife Protection Act, 1972.

Buffer Zone

An area surrounding a core protected area where regulated human activities, eco-tourism, and conservation awareness programs are permitted.

Project Elephant

A Centrally Sponsored Scheme launched in 1992 to protect wild elephants, their habitats, and migration corridors across India.

Elephant Corridor

A narrow strip of land that connects two larger elephant habitats, allowing free movement of elephant herds and maintaining genetic diversity.

Prelims: how it could come

With reference to the Asian Elephant (*Elephas maximus*), consider the following statements:

- 1. It is listed as Vulnerable on the International Union for Conservation of Nature (IUCN) Red List.**
- 2. It is classified under Schedule II of the Wildlife Protection Act, 1972.**

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Answer: (d) Statement 1 is incorrect: The Asian Elephant is listed as Endangered on the International Union for Conservation of Nature (IUCN) Red List, not Vulnerable. Statement 2 is incorrect: Asian Elephants are protected under Schedule I of the Wildlife Protection Act, 1972, providing them the highest level of legal protection in India.

Mains: how it could come

Examine the role of buffer zone management and captive elephant care in strengthening wildlife conservation efforts in India. (150 words, 10 marks)

Buffer zones act as transition regions between human settlements and core wildlife habitats, absorbing ecological shocks and fostering peaceful co-existence.

- **Buffer Zone Functions:** Reduces edge effects on core zones, accommodates regulated eco-tourism, and offers sustainable livelihood options to peripheral communities.
- **Captive Wildlife Care:** Health rejuvenation camps for forest service elephants enhance operational efficiency for patrolling inaccessible core terrains.
- **Conflict Mitigation:** Engaging local populations during events in buffer zones builds public empathy and lowers human-elephant conflict instances.
- **Ecological Corridors:** Effective buffer management ensures structural connectivity for migratory wildlife like elephant herds.

Strengthening Eco-Development Committees and promoting participatory management are vital for sustaining long-term biodiversity conservation.

Keywords to use: Buffer Zone Management, Captive Elephant Welfare, Human-Wildlife Conflict, Eco-Development Committees, Project Elephant

Newspaper words

advocated	समर्थन किया	publicly recommended or supported a policy or cause
multilateral	बहुपक्षीय	involving three or more nations or groups
attribution	उत्तरदायित्व निर्धारण	the action of identifying the cause or responsible party for an event
accountability	जवाबदेही	the fact of being responsible for one's decisions or actions
rejuvenation	कायाकल्प	the process of restoring health, vitality, or energy